

vieme:

$$f((2,1,0,1)^T) = (4,1,2,5)^T$$

$$f((1,0,1,2)^T) = (2,2,-2,-2)^T$$

$$V = \text{span} \{ (2,1,0,1)^T; (1,0,1,2)^T \}$$

1. nájdime maticu operátora f vzhľadom k bázi B

$$B = ((2,1,0,1)^T, (1,0,1,2)^T)$$

$$\hookrightarrow a \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}$$

$$\hookrightarrow \text{vidíme} \dots \quad a=1$$

$$b=2$$

$$\hookrightarrow c \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + d \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

$$\hookrightarrow \text{vidíme} \dots \quad c=2$$

$$d=-2$$

$$[f]_B = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

2. diagonalizujeme maticu $[f]_B$

$\hookrightarrow$ chceme nájsť jej vlastné čísla

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = (1-\lambda)(-2-\lambda)-4 = \lambda^2+\lambda-6 = (\lambda+3)(\lambda-2)=0$$

$$\hookrightarrow \lambda_1 = 2 \quad \lambda_2 = -3$$

$\hookrightarrow$ hledáme vlastní vektory v síř. bázi B
 $\lambda_1 \dots \text{Ker} \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix}$

$$\begin{pmatrix} -1 & 2 & | & 0 \\ 2 & -4 & | & 0 \end{pmatrix} \sim \begin{pmatrix} -1 & 2 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix} \quad \text{čiře } [\vec{v}_1]_B = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\lambda_2 \dots \text{Ker} \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 2 & | & 0 \\ 2 & 1 & | & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 0 & | & 0 \\ 2 & 1 & | & 0 \end{pmatrix} \quad \text{čiře } [\vec{v}_2]_B = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$2 \text{ řeho } [f]_B = R A R^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}^*$$

$$3. f^m (5, 1, 3, 7)^T$$

$\hookrightarrow$ vyjádřime v bázi B vektor $\vec{w} = \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}$

$$e \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + f \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}$$

$$\hookrightarrow \begin{matrix} e=1 \\ f=3 \end{matrix} \rightarrow [\vec{w}]_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$[f^n(\vec{w})]_B = [f]_B^n [\vec{w}]_B = R A^n R^{-1} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

↳ prechod z bázy B do bázy $C = (\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \end{pmatrix})$

↳ umocnenie

↳ prechod z bázy C do bázy B

1. $R^{-1} [\vec{w}]_B$

$$\frac{1}{5} \begin{pmatrix} 2 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \frac{1}{5} \left(\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} -1 \\ -2 \end{pmatrix} \right) = \frac{1}{5} \begin{pmatrix} -1 \\ -5 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} = [\vec{w}]_C$$

2. $A^n R^{-1} [\vec{w}]_B$

$$A^n = \begin{pmatrix} 2^n & 0 \\ 0 & -3^n \end{pmatrix}$$

$$A^n [\vec{w}]_C = \begin{pmatrix} 2^n & 0 \\ 0 & -3^n \end{pmatrix} \begin{pmatrix} -1 \\ -1 \end{pmatrix} = \begin{pmatrix} -2^n \\ -(-3)^n \end{pmatrix}$$

3. $R A^n R^{-1} [\vec{w}]_B$

$$\begin{pmatrix} 2 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} -2^n \\ -(-3)^n \end{pmatrix} = 2^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} - (-3)^n \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \cdot 2^n - (-3)^n \\ 2^n + 2(-3)^n \end{pmatrix}$$

h. převedně spät

$$f^n(\vec{w}) = (2 \cdot 2^n - (-3)^n) \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + (2^n + 2(-3)^n) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} =$$

$$= \begin{bmatrix} 5 \cdot 2^n \\ 2 \cdot 2^n - (-3)^n \\ 2^n + 2(-3)^n \\ 4 \cdot 2^n + 3 \cdot (-3)^n \end{bmatrix}$$

$$* R^{-1} = \frac{1}{\det R} \operatorname{adj}(R) = \frac{1}{5} \begin{pmatrix} -2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$